MATH 111 Calculus 1

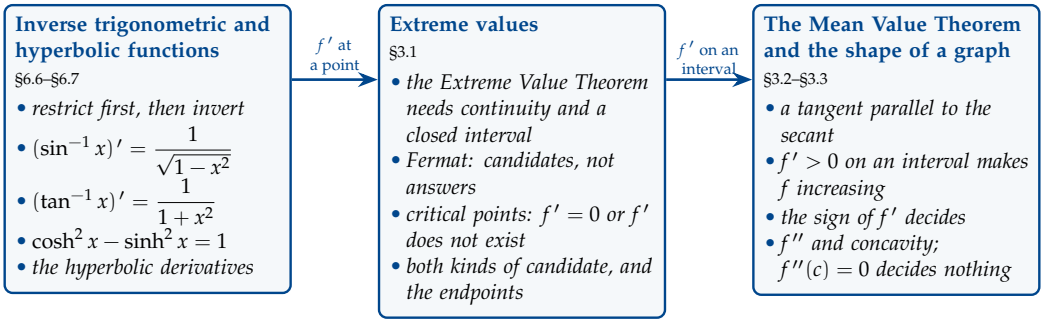
Revision

From the inverse sine to the shape of a graph — in questions

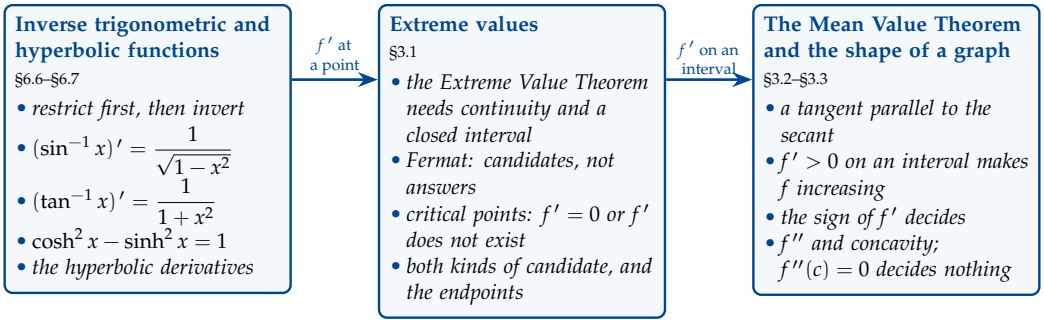
Week 7 · Lecture 19 · §6.6–§6.7, §3.1–§3.3

Dr. Salnikova Chekeres · New Uzbekistan University · Fall 2026

The map — §6.6–§6.7 and §3.1–§3.3 on one page



The map — §6.6–§6.7 and §3.1–§3.3 on one page



Today

Three boxes, one tool — the derivative: computed for the new functions; read at a point, where it gives only *candidates* (Fermat); read on an interval, where it gives the *shape* of the graph (the Mean Value Theorem) — and every theorem only with its hypotheses.

STAND UP

"If f is continuous on $[a, b]$, then f takes every value between its absolute minimum value and its absolute maximum value on $[a, b]$."

Stand up if you think this is TRUE.

Stay standing until I say "sit". Then find someone with the other opinion — 60 seconds.

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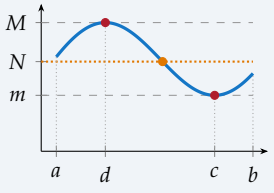
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Stand up if you think this is TRUE.

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Verdict

TRUE — two theorems together. The Extreme Value Theorem: $f(c) = m$ and $f(d) = M$ for some c, d in $[a, b]$. The Intermediate Value Theorem on the interval between c and d: every N between m and M is a value of f. Both hypotheses matter: x on $(0, 1)$ has no largest or smallest value; a step from 0 to 1 on $[0, 2]$ never takes $1/2$.



Question 61

Let $f(x) = \tan^{-1}\left(\dfrac{1}{x^2}\right)$ for $x \neq 0$, and $f(0) = \dfrac{\pi}{2}$. At $x = 0$ the function f is

A. not continuous: $1/x^2$ is not defined at 0.

B. not continuous: a jump, from $-\pi/2$ to $\pi/2$.

C. continuous.

D. not continuous: an infinite discontinuity, because $\lim_{x\to 0} 1/x^2 = \infty$.

In your team

Write your own letter first — then talk. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

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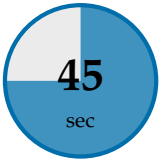
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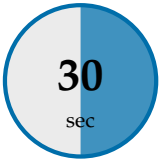
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Hands down. Thank you.

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ANSWER

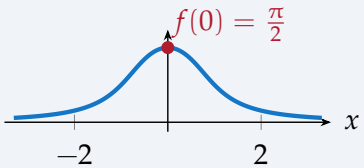
Answer

C. $\lim_{x\to 0} \frac{1}{x^2} = \infty$ from both sides; with $u = \frac{1}{x^2}$,
$\lim_{x\to 0} f(x) = \lim_{u\to\infty} \tan^{-1} u = \frac{\pi}{2} = f(0)$.

A: the formula is not used at 0; $f(0)$ is given.

B is $\tan^{-1} \frac{1}{x}$ ($\frac{1}{x} < 0$ on the left).

D: $\tan^{-1}$ of a large number is close to $\frac{\pi}{2}$.



for $x \neq 0$: $\tan^{-1} \frac{1}{x^2} = \frac{\pi}{2} - \tan^{-1} x^2$

Question 62

$$\frac{d}{dx}\left(x\cosh x - \sinh x\right) = ?$$

A. $x\sinh x$

B. $-x\sinh x$

C. $\sinh x - \cosh x$

D. 0

In your team

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THINK & TALK · 1 minute

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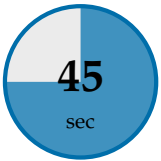
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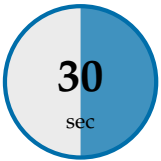
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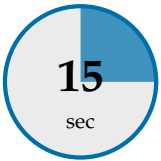
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Hands down. Thank you.

A B C D

Answer

A. The product rule: $(x\cosh x)' = \cosh x + x\sinh x$ — $(\cosh x)' = \sinh x$, no minus sign; and $(\sinh x)' = \cosh x$. So the derivative is $\cosh x + x\sinh x - \cosh x = x\sinh x$.

B gives cosh the cosine's minus sign.

C multiplies the derivatives of the factors: $1 \cdot \sinh x$.

D is half a product rule: $\cosh x$ alone, and everything cancels.

STAND UP

"If f is continuous on $[a, b]$ and not constant, then f has a critical point in (a, b)."

Stand up if you think this is TRUE.

Stay standing until I say "sit". Then find someone with the other opinion — 60 seconds.

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"If f is continuous on $[a, b]$ and not constant, then f has a critical point in (a, b)."

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Verdict

FALSE. $f(x) = x$ on $[0, 1]$: $f'(x) = 1$ for every x — no critical point at all. Its minimum $f(0) = 0$ and its maximum $f(1) = 1$ are at the endpoints. The Extreme Value Theorem promises extreme *values*, not critical points: that is why the endpoints are candidates.

Question 63

What are the critical points of $f(x) = 3\sqrt[3]{x^2} - 2x$?

A. 1 only

B. 0 and $\frac{27}{8}$

C. $\frac{16}{81}$ only

D. 0 and 1

In your team

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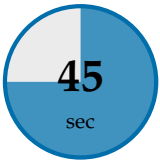
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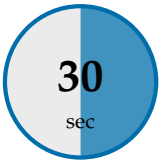
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A B C D

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Hands down. Thank you.

A B C D

ANSWER

Answer

D. For $x \neq 0$: $f'(x) = 2x^{-1/3} - 2 = \dfrac{2\left(1 - \sqrt[3]{x}\right)}{\sqrt[3]{x}}$, zero at $x = 1$. At 0: $\lim\limits_{h \to 0} \dfrac{f(h) - f(0)}{h} = \lim\limits_{h \to 0} \left(\dfrac{3}{\sqrt[3]{h}} - 2 \right)$ does not exist (a cusp), and 0 is in the domain: a critical point.

A forgets where f' does not exist.

B: the zeros of f, not of f'.

C reads $\sqrt[3]{x^2}$ as $x^{3/2}$; it is $x^{2/3}$.

Question 64

The absolute maximum value of $f(x) = x^3 + 3x^2 - 24x - 4$ on the interval $[1, 4]$ is

A. 76

B. 12

C. 32

D. 4

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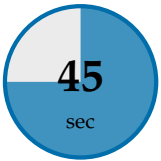


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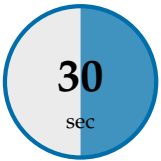
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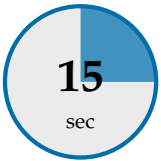
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In your team

Write your own letter first — then talk. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

THINK & TALK · 1 minute

The absolute maximum value of $f(x) = x^3 + 3x^2 - 24x - 4$ on the interval $[1, 4]$ is

A. 76

B. 12

C. 32

D. 4



In your team

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Hands down. Thank you.

A B C D

ANSWER

Answer

B. The Closed Interval Method. $f'(x) = 3x^2 + 6x - 24 = 3(x+4)(x-2)$: critical points -4 and 2, and only 2 lies in $[1, 4]$: $f(2) = -32$. Endpoints: $f(1) = -24$, $f(4) = 12$. The largest is $f(4) = 12$ — at an endpoint.

A is $f(-4) = 76$: -4 is not in $[1, 4]$, so it is not a candidate.

C reads "absolute" as "absolute value": $|f(2)| = 32$, but $f(2) = -32$ is the absolute *minimum*.

D gives where the maximum is, not its value.

STAND UP

"If $f'(x) > 0$ for every x in the domain of f, then f is increasing on its domain."

Stand up if you think this is TRUE.

Stay standing until I say "sit". Then find someone with the other opinion — 60 seconds.

STAND UP

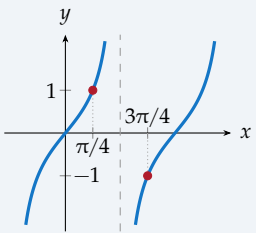
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Stand up if you think this is TRUE.

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Verdict

FALSE. $f(x) = \tan x$: $f'(x) = \sec^2 x > 0$ at every point of its domain, but $\tan(\pi/4) = 1 > -1 = \tan(3\pi/4)$, although $\pi/4 < 3\pi/4$. The test is about an interval: tan is increasing on $(-\pi/2, \pi/2)$ and on $(\pi/2, 3\pi/2)$, separately. Its proof uses the Mean Value Theorem on the whole interval between two points — and between $\pi/4$ and $3\pi/4$ lies $\pi/2$.



Question 65

f is differentiable on $\mathbb{R}$, $f'(x) \leq 2$ for every x, $f(0) = 0$ and $f(4) = 8$.
What can be said about $f(2)$?

A. $f(2) \leq 4$, and nothing more.

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C. Nothing: the Mean Value Theorem gives a number c, not a value of f.

D. $f(2) = 4$ exactly.

In your team

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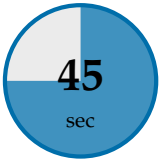
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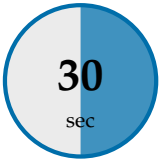
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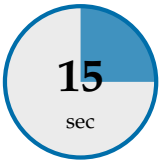
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ANSWER

Answer

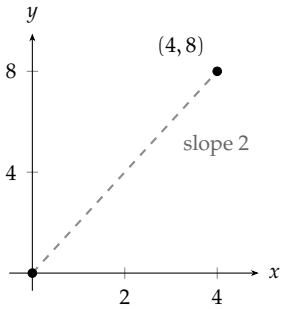
D. The Mean Value Theorem, twice. On $[0,2]$: $f(2)-f(0)=f'(c)\cdot 2\le 4$, so $f(2)\le 4$. On $[2,4]$: $f(4)-f(2)=f'(d)\cdot 2\le 4$, so $f(2)\ge 8-4=4$. Together: $f(2)=4$.

A and B each use one half of the interval.

C: the theorem names no c — but $f'\le 2$ turns "some c" into a bound on f, and here two bounds meet.

Question 65 — the argument, written out

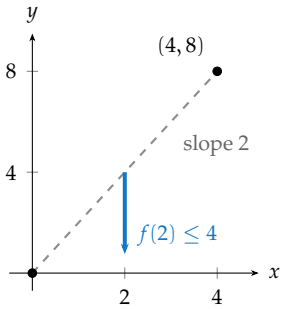
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On $[0, 2]$: there is a number c in $(0, 2)$ with $f(2) - f(0) = f'(c)\,(2 - 0)$, so $f(2) = 2f'(c) \leq 4$.

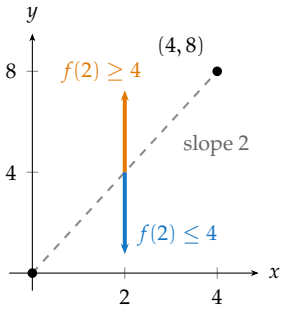


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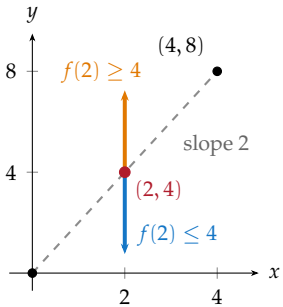
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Hence $f(2)=4$. ■



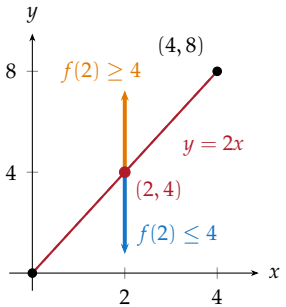
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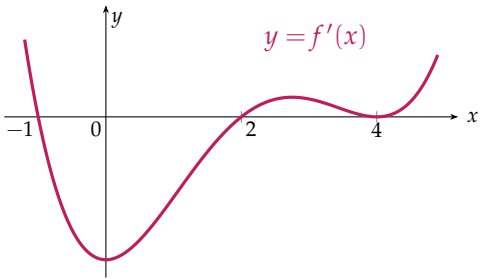
Hence $f(2)=4$. ∎



For every x in $[0,4]$ the same two steps give $f(x)\le 2x$ and $8-f(x)\le 2(4-x)$; so $f(x)=2x$: the graph is the line through $(0,0)$ and $(4,8)$. A slope never above 2 and a rise of 8 over 4 leave no room.

Question 66

The graph of the *derivative* f' of a function f is shown. At which x does f have a local minimum?



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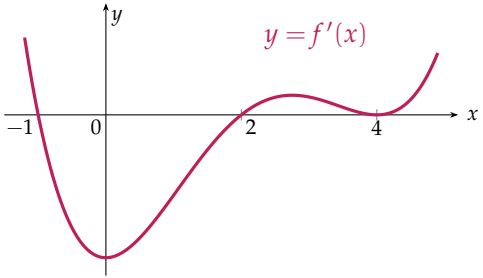
C. $x = 2$ **D.** $x = 4$

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THINK & TALK · 1 minute

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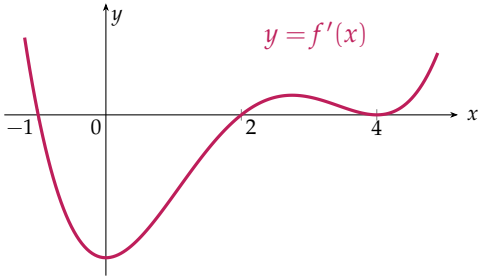
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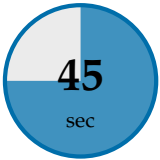
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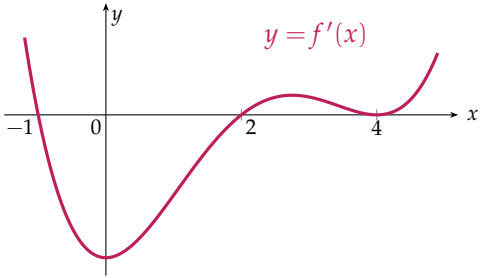


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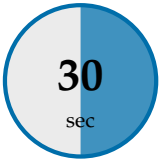
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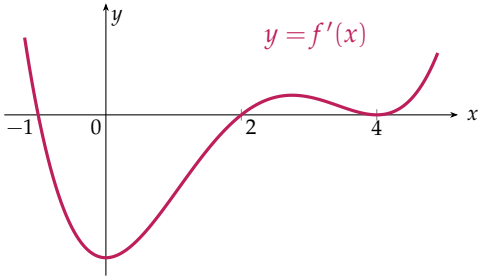


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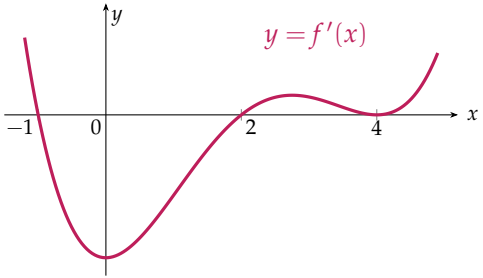


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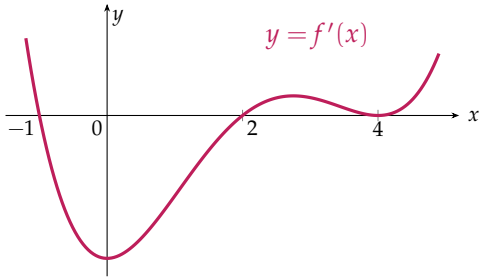


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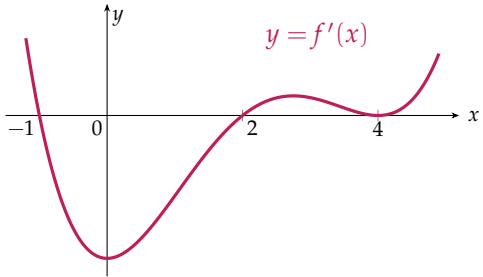
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A B C D

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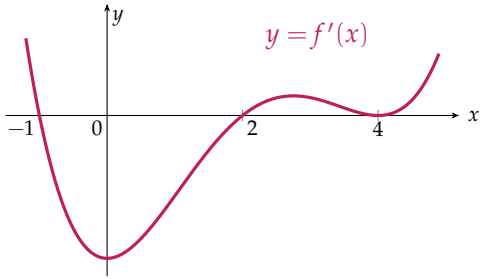
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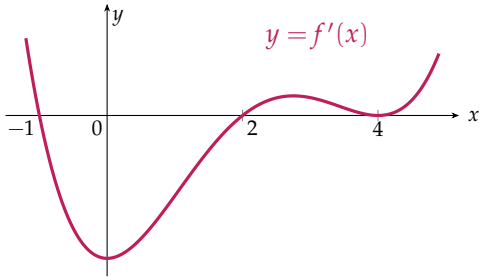
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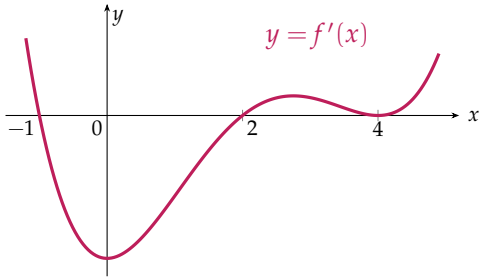
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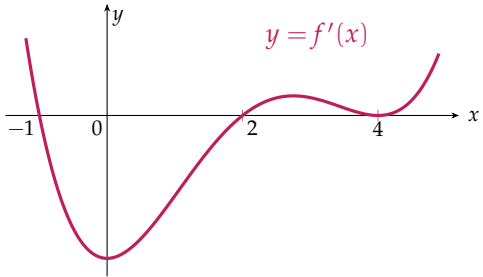
C. $x = 2$ **D.** $x = 4$

Eyes on the screen. Hands up when your letter appears — hold for three seconds.

A B C **D**

VOTE

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C. $x = 2$ **D.** $x = 4$

Hands down. Thank you.

A B C D

Answer

C. f' is the *slope* of f: at $x = 2$ it changes from $-$ to $+$, so f falls, then rises — a local minimum (the First Derivative Test).

A: from $+$ to $-$ at -1, a local *maximum*.

B reads the curve as the graph of f: at 0, f falls fastest.

D: $f'(4) = 0$, but $f' > 0$ on both sides.

Question 67

True or false ?

If $f''(x) > 0$ for every x, then f has at most one critical point.

In your team

Decide your own answer first — true or false — then talk. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

THINK & TALK · 40 seconds

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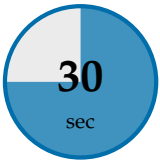
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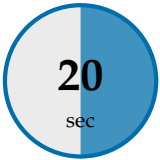
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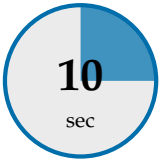
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VOTE

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If $f''(x) > 0$ for every x, then f has at most one critical point.

Eyes on the screen. Hands up when your answer appears — hold for three seconds.

TRUE FALSE

VOTE

True or false ?

If $f''(x) > 0$ for every x, then f has at most one critical point.

Eyes on the screen. Hands up when your answer appears — hold for three seconds.

TRUE

FALSE

VOTE

True or false ?

If $f''(x) > 0$ for every x, then f has at most one critical point.

Eyes on the screen. Hands up when your answer appears — hold for three seconds.

TRUE

FALSE

VOTE

True or false ?

If $f''(x) > 0$ for every x, then f has at most one critical point.

Hands down. Thank you.

TRUE FALSE

ANSWER

Answer

True. PROOF f'' exists everywhere, so f is differentiable everywhere: every critical point is a zero of f'. $f'' > 0$ on $\mathbb{R}$, so f' is increasing on $\mathbb{R}$ (the Increasing/Decreasing Test, applied to f'): f' takes the value 0 at most once. ■

Examples: e^x (no critical point), x^2 (one, at 0).

Question 68

$f(x) = x^3 + 6x^2 - 15x$. What is the largest open interval on which the graph of f is concave downward ?

A. $(-\infty, -2)$

B. $(-2, \infty)$

C. $(-5, 1)$

D. $(-5, -2)$

In your team

Write your own letter first — then talk. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

THINK & TALK · 1 minute

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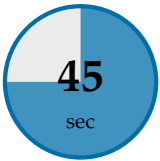
60 sec

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THINK & TALK · 1 minute

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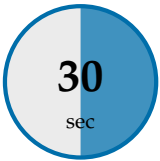
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Hands down. Thank you.

A B C D

ANSWER

Answer

A. Concavity is decided by f'': $f'(x) = 3x^2 + 12x - 15 = 3(x+5)(x-1)$, $f''(x) = 6x + 12$, negative exactly for $x < -2$; inflection point $(-2, 46)$.

B reverses the sign.

C is where f is *decreasing* ($f' < 0$): on $(-2, 1)$ the graph is decreasing and concave *upward*.

D asks for $f' < 0$ as well — concavity asks only about f''; the graph is concave downward on $(-5, -2)$ too, but that is not the largest such interval.

By heart (1) — the theorems, with their hypotheses

Extreme values

EVT: f continuous on $[a, b] \Rightarrow$ absolute maximum and minimum attained.

Fermat: local extremum at c and $f'(c)$ exists $\Rightarrow f'(c) = 0$.

Critical point: $f'(c) = 0$ or $f'(c)$ does not exist.

Closed Interval Method: the critical points in (a, b), and a, b.

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The Mean Value Theorem and the shape

MVT: f continuous on $[a, b]$, differentiable on $(a, b) \Rightarrow f(b) - f(a) = f'(c)\,(b - a)$.

I/D: $f' > 0$ on an *interval* $\Rightarrow$ increasing there.

First Derivative Test: f' from $-$ to $+$: minimum; from $+$ to $-$: maximum.

$f'' > 0$: concave upward; inflection: f'' changes sign.

Second Derivative Test: $f''(c) = 0$: no information.

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The habits

Check both kinds of candidate, and the endpoints. State the hypotheses before you use a theorem. An interval, not just a domain. $f''(c) = 0$ *decides nothing.*

Inverse trigonometric functions — restrict first, then invert

$\sin^{-1} x$: domain $[-1,1]$, range $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$; $\dfrac{d}{dx}\sin^{-1} x = \dfrac{1}{\sqrt{1-x^2}}$ $(-1 < x < 1)$

$\cos^{-1} x$: domain $[-1,1]$, range $[0, \pi]$; $\dfrac{d}{dx}\cos^{-1} x = -\dfrac{1}{\sqrt{1-x^2}}$ $(-1 < x < 1)$

$\tan^{-1} x$: domain $\mathbb{R}$, range $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$; $\dfrac{d}{dx}\tan^{-1} x = \dfrac{1}{1+x^2}$; asymptotes $y = \pm\frac{\pi}{2}$

By heart (2) — the new functions

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Hyperbolic functions — no minus sign

$$\sinh x = \frac{e^x - e^{-x}}{2}, \quad \cosh x = \frac{e^x + e^{-x}}{2}, \quad \tanh x = \frac{\sinh x}{\cosh x}; \qquad \cosh^2 x - \sinh^2 x = 1.$$

$$\frac{d}{dx}\sinh x = \cosh x, \qquad \frac{d}{dx}\cosh x = \sinh x \text{ (no minus sign)}, \qquad \frac{d}{dx}\tanh x = \operatorname{sech}^2 x.$$

Inflection points — what happens to the tangent?

STAND UP

"At an inflection point, the tangent line is horizontal."

Stand up if you think this is TRUE.

Stay standing until I say "sit". Then find someone with the other opinion — 60 seconds.

STAND UP

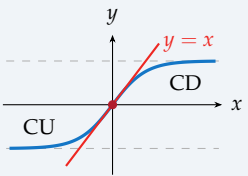
"At an inflection point, the tangent line is horizontal."

Stand up if you think this is TRUE.

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Verdict

FALSE. $f(x) = \tanh x$: $f'(x) = \operatorname{sech}^2 x = 1 - \tanh^2 x$, so $f''(x) = -2\tanh x\,(1 - \tanh^2 x)$: positive for $x < 0$, negative for $x > 0$. So $(0, 0)$ is an inflection point — and the tangent there is $y = x$, of slope $f'(0) = 1$. An inflection point is where the concavity changes; a horizontal tangent there (as for x^3 at 0) is a special case.



Before the examination — three things

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Next:

the midterm examination.

EXIT TICKET

Tear a quarter page from a notebook — one per team.

1. Team name; names of those present today.
2. One sentence: *the one kind of question from the whole course that I still cannot do is … (a kind of question, not a chapter).*
3. Your muddiest point — the one thing still unclear.
4. One moment today that surprised you.

Give it to the professor or the course assistant as you go. Thank you.